

From the slender-jet hierarchy to a uniform-filament force balance

Pre-requisites: *Viscoelastic slender jets* and *Viscoelastic slender jets at $O(\varepsilon^2)$* . Primes are ∂_z , dots are ∂_t .

Elastocapillary thinning of a filament is usually analysed not with the one-dimensional slender-jet equations but with a force balance across a single cross-section: the axial tension F_z transmitted along the filament equals the integrated axial stress plus the surface-tension line force. This is the language of Entov and Hinch (1997), of the beads-on-a-string analysis of Clasen et al. (2006), and of the ROJER and CaBER analyses of Mathues et al. (2018). The purpose of this note is to show how that force balance follows from the slender-jet hierarchy, which terms of the $O(\varepsilon^2)$ system it keeps, and what the inertial correction in it must be.

The ansatz

Take a locally cylindrical filament of radius $h(t)$ in homogeneous uniaxial extension:

$$h = h(t), \quad v_z = v_0(z, t) = -2\frac{\dot{h}}{h}z, \quad v_r = \frac{\dot{h}}{h}r.$$

The radial velocity is the exact consequence of continuity, $v_r = -\frac{r}{2}v'_0$, and the strain rate is $\dot{\epsilon} \equiv v'_0 = -2\dot{h}/h$. Every z -derivative of h and every second z -derivative of v_0 vanishes.

What the $O(\varepsilon^2)$ system reduces to

Insert the ansatz into the second-order closures of *the previous note*:

- (T0) with $h' = v''_0 = 0$ and no polymer shear stress gives $v_2 = 0$. The axial velocity is exactly plug-like.
- (T2) then gives $v_4 = 0$.
- (C) is satisfied exactly at every order: $\partial_t(h^2) + \partial_z(h^2v_0) = 2h\dot{h} - 2h\dot{h} = 0$.
- (R) collapses to two inertial terms. With $v'_0 = -2\dot{h}/h$,

$$\frac{\rho}{4}\partial_t v'_0 = -\frac{\rho}{2}\left(\frac{\ddot{h}}{h} - \frac{\dot{h}^2}{h^2}\right), \quad -\frac{\rho}{8}(v'_0)^2 = -\frac{\rho}{2}\frac{\dot{h}^2}{h^2},$$

so the $\dot{h}^2$ contributions cancel and

$$p_2 = -\frac{\rho}{2}\frac{\ddot{h}}{h}, \quad p(r, t) = p_0 - \frac{\rho}{2}\frac{\ddot{h}}{h}r^2.$$

- (B) keeps only the pressure term: $\mathcal{B} = -h^2p_2 = \frac{\rho}{2}h\ddot{h}$.

So under this ansatz the whole second-order system reduces to one statement: the pressure is parabolic in r , with curvature set by the radial acceleration $\ddot{h}$. This is exactly what the radial momentum equation says directly, since the radial acceleration of a fluid particle is

$$a_r = \partial_t v_r + v_r \partial_r v_r = \left(\frac{\ddot{h}}{h} - \frac{\dot{h}^2}{h^2}\right)r + \frac{\dot{h}^2}{h^2}r = \frac{\ddot{h}}{h}r,$$

and $\partial_r p = -\rho a_r$ integrates to the parabola above. The cancellation of $\dot{h}^2$ between the local and convective accelerations is the whole point: keeping only the convective piece would give $p_2 = -\rho\dot{h}^2/(2h^2)$ and an inertial correction proportional to $\rho\dot{h}^2/2$, which is the Bernoulli-type form that appears in several filament-thinning analyses. It is not the consistent second-order term.

Three pressures

Three pressures appear in the force balance, and they differ at $O(\varepsilon^2)$:

$$p_0 = p(0, t), \quad p_h = p(h, t) = p_0 - \frac{\rho}{2} h \ddot{h}, \quad \bar{p} = \frac{1}{\pi h^2} \int_0^h p \, 2\pi r \, dr = p_0 - \frac{\rho}{4} h \ddot{h}.$$

The leading-order slender-jet equation (M0) uses p_0 . The traction condition fixes p_h . The force across the cross-section needs $\bar{p}$. Eliminating p_0 ,

$$\bar{p} = p_h + \frac{\rho}{4} h \ddot{h}.$$

Normal traction and the axial force

The normal traction at $r = h$ with $n = e_r$ and $\kappa = 1/h$:

$$-p_h + 2\eta \partial_r v_r + \sigma_{p,rr} = -\frac{\gamma}{h} \Rightarrow p_h = \frac{\gamma}{h} + 2\eta \frac{\dot{h}}{h} + \Sigma_{rr}.$$

The axial force through a cross-section $\mathcal{S}$ of the filament is the integrated axial stress plus the surface-tension line force acting on the circumference:

$$F_z = \int_{\mathcal{S}} (-p + 2\eta \partial_z v_z + \sigma_{p,zz}) \, dS + 2\pi\gamma h = \pi h^2 \left(-\bar{p} - 4\eta \frac{\dot{h}}{h} + \Sigma_{zz} \right) + 2\pi\gamma h.$$

Substituting $\bar{p}$:

$$\boxed{\frac{F_z}{\pi h^2} = \frac{\gamma}{h} - 6\eta \frac{\dot{h}}{h} + (\Sigma_{zz} - \Sigma_{rr}) - \frac{\rho}{4} h \ddot{h}.}$$

The three familiar terms are the capillary pressure (the line force $2\gamma/h$ minus the Laplace pressure γ/h), the Trouton-ratio viscous stress $3\eta\dot{\epsilon}$, and the polymer normal-stress difference. The fourth is the radial-inertia correction. It is $O(\varepsilon^2)$ relative to the others and is the only second-order term that a uniform filament retains.

Why the momentum equation is not used directly: With the ansatz, every z -gradient on the right-hand side of (E2) vanishes and the axial equation reduces to $\rho(\partial_t v_0 + v_0 v_0') = 0$, which the homogeneous-extension velocity does not satisfy (it would require $h\ddot{h} = 3\dot{h}^2$). The axial inertia of the stretching flow is balanced by the z -variation of F_z along the filament, which a single-radius model does not resolve. The force balance therefore treats F_z as an imposed tension, set by whatever the filament is attached to: end plates in a rheometer, a nozzle, or the base of a jet. This is the price of the reduction, and it is why the choice of F_z is a modelling closure rather than a derived quantity.

Polymer stresses

For homogeneous extension with r -uniform polymer stress, the second-order polymer coefficients $\Sigma_{rr,2}$, Θ_2 , S and S_3 all vanish, so the polymer enters the force balance only through $\Delta\Sigma = \Sigma_{zz} - \Sigma_{rr}$. For an Oldroyd-B liquid, $\sigma_p = G(\mathbb{A} - \mathbb{I})$, and in uniaxial extension the axial conformation obeys

$$\frac{dA_{zz}}{dt} = 2\dot{\epsilon}A_{zz} - \frac{A_{zz} - 1}{\lambda} = -4\frac{\dot{h}}{h}A_{zz} - \frac{A_{zz} - 1}{\lambda},$$

whose strongly stretched solution, starting from a relaxed state at radius h_0 , is $A_{zz} \simeq (h_0/h)^4 \exp(-t/\lambda)$. With $A_{zz} \gg A_{rr}$ the normal-stress difference is $\Delta\Sigma \simeq GA_{zz}$. Substituting gives the reduced equation

$$\frac{F_z}{\pi h^2} = \frac{\gamma}{h} - 6\eta \frac{\dot{h}}{h} - \frac{\rho}{4} h \ddot{h} + G \left(\frac{h_0}{h} \right)^4 e^{-t/\lambda}.$$

When the capillary term dominates the tension, $F_z/(\pi h^2) \ll \gamma/h$, this recovers the classical exponential thinning $h \propto \exp(-t/3\lambda)$ of Entov and Hinch (1997). When the tension is imposed and dominates, $F_z = \pi h^2 G(h_0/h)^4 e^{-t/\lambda}$ gives instead $h \propto \exp(-t/2\lambda)$. For an exponentially thinning filament the inertial correction scales as $\rho h^2/\lambda^2$ and disappears as $h \rightarrow 0$, so it only matters during transients in which $\ddot{h}$ is not slaved to $\dot{h}^2/h$. That is precisely the regime of a freshly formed jet, and it is the form used for the transient stretching of elastocapillary Worthington jets in Sen, Sanjay et al. (in preparation).

Summary of the bridge

Slender-jet hierarchy	Uniform-filament force balance
leading-order momentum (M0), uses p_0	cross-section force, uses $\bar{p}$
normal traction fixes p_0 at $O(1)$	normal traction fixes p_h
radial equation (R) fixes p_2	$p_2 = -\rho \dot{h}/(2h)$
$\mathcal{B} = -h^2 p_2$ at $O(\varepsilon^2)$	$\bar{p} - p_h = \rho h \ddot{h}/4$
$-\partial_z \mathcal{B}$ forces v_0	$-\rho h \ddot{h}/4$ in $F_z/(\pi h^2)$
v_2, v_4 , curvature and slope corrections	all identically zero

• Related

- *Viscoelastic slender jets (leading order)*
- *Viscoelastic slender jets at $O(\varepsilon^2)$*
- *continuity-slender-jet*
- *dynamic-BC-slender-jet*